

Exam 2 calculation component

1. a) Stump 1

	X_1	X_2	Y	Weight 1 (w_1)	Predicts (y_1)
0	1	2	1	$0.2 = 1/5$	1
1	2	5	1	0.2	1
2	3	4	-1	0.2	-1
3	4	0	-1	0.2	-1
4	0	1	-1	0.2	1

3 stumps: $1(X_1 < 2.5)$ $L=1$

$1(X_2 > 1.5)$

$1(X_2 > 4.5)$

b) decrease weights for correctly classified observations $\rightarrow 0, 1, 2, 3$
 increase weights for misclassified observations $\rightarrow 4$
 - (based on Stump 1)

c) errors of Stump 1 = $\sum$ misclassified weights

$$E_1 = 0.2$$

$$\alpha_1 = L \cdot \frac{1}{2} \cdot \ln\left(\frac{1-E_1}{E_1}\right) = 1\left(\frac{1}{2}\right) \cdot \ln\left(\frac{0.8}{0.2}\right) = 0.6931$$

for misclassified:

$$w_{\text{new}} = w_{\text{old}} \cdot e^{\alpha}$$

$$= 0.2 \cdot e^{0.6931} = 0.4$$

for correctly classified:

$$w_{\text{new}} = w_{\text{old}} \cdot e^{-\alpha}$$

$$= 0.2 \cdot e^{-0.6931} = 0.1$$

normalize by dividing by: $(0.4 \cdot 1 + 0.1 \times 4) = 0.8$

	X_1	X_2	Y	w_1	y_1	w_2	y_2	w_3	y_3
0	1	2	1	0.2	1	0.125	1	0.0714	-1
1	2	5	1	0.2	1	0.125	1	0.0714	1
2	3	4	-1	0.2	-1	0.125	1	0.5	-1
3	4	0	-1	0.2	-1	0.125	-1	0.0714	-1
4	0	1	-1	0.2	1	0.5	-1	0.2857	-1

errors of stump 2 = $\sum$ misclassified $\rightarrow E_2 = 0.125$

$$\alpha_2 = L \cdot \frac{1}{2} \cdot \ln\left(\frac{1-E_2}{E_2}\right) = 1\left(\frac{1}{2}\right) \cdot \ln\left(\frac{0.875}{0.125}\right) = 0.97296$$

for misclassified:

$$w_{\text{new}} = w_{\text{old}} \cdot e^{\alpha}$$

$$= 0.125 \cdot e^{0.97296} = 0.3307189$$

for correctly classified

$$w_{\text{new}} = w_{\text{old}} \cdot e^{-\alpha}$$

$$= 0.125 \cdot e^{-0.97296} = 0.0472$$

$$= 0.5 \cdot e^{-0.97296} = 0.18898$$

normalize by dividing by: $(0.33 \cdot 1 + 0.18898 \cdot 1 + 0.0472 \cdot 3) = 0.6614378$

errors of stump 3 = $\sum$ misclassified $\rightarrow \boxed{\epsilon_3 = 0.0714}$

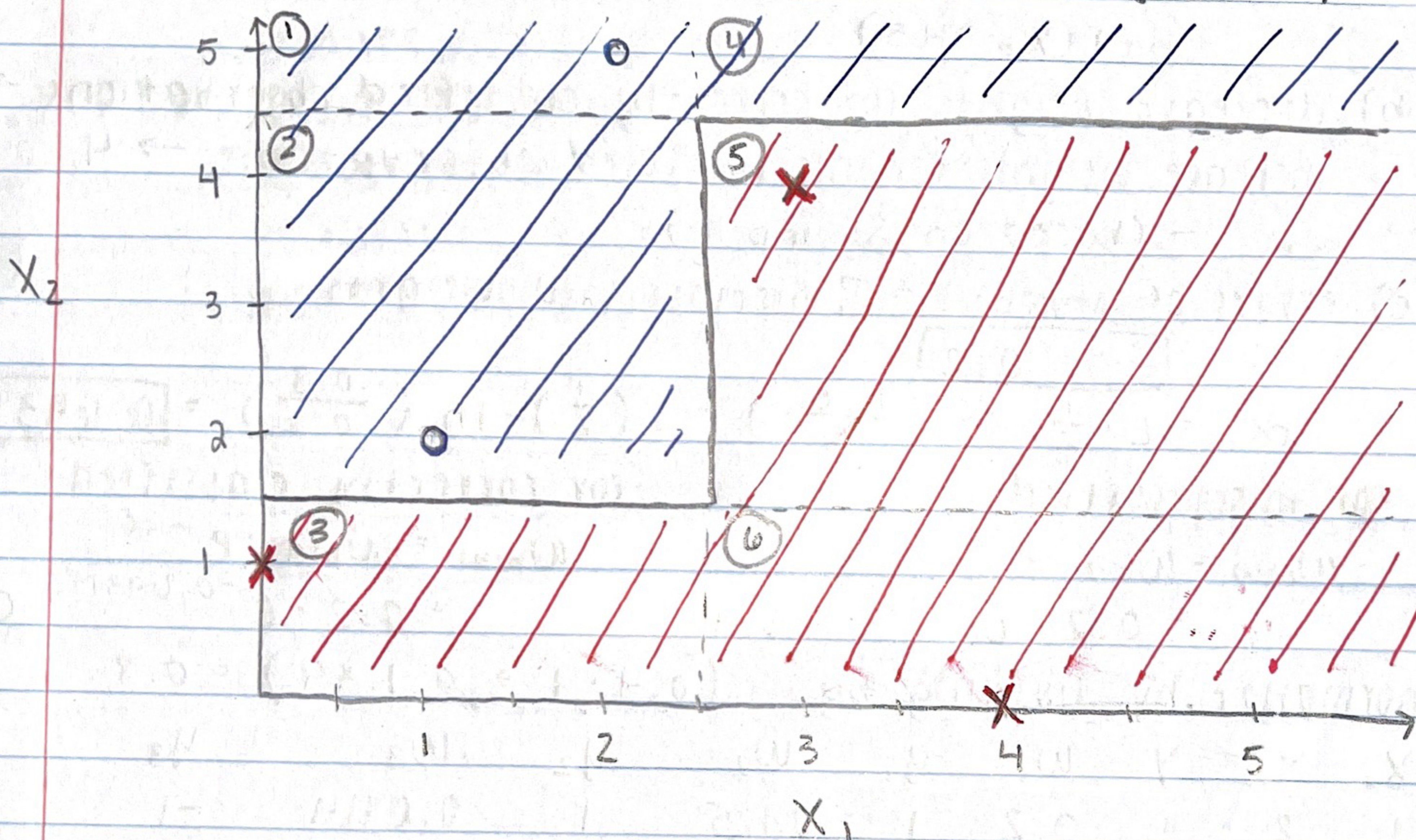
$$\alpha_3 = L \cdot \frac{1}{2} \cdot \ln\left(\frac{1-\epsilon_3}{\epsilon_3}\right) = 1 \cdot \frac{1}{2} \cdot \ln\left(\frac{1-0.0714}{0.0714}\right) = \boxed{1.2825}$$

d) Voting power = $\alpha \rightarrow \alpha_1 = 0.6931$
 $\alpha_2 = 0.97296$
 $\alpha_3 = 1.2825$

Stump 3 has the highest voting power

e) Prediction function of adaboost:

$$H(x) = \text{Sign}(0.6931 \cdot I(x_1 < 2.5) + 0.97296 \cdot I(x_2 > 1.5) + 1.2825 \cdot I(x_2 > 4.5))$$



$y = -1$

$y = 1$

Section:

1. $\text{sign}(.6931(-1) + 0.973(1) + 1.2827(1)) = \text{positive}$

2. $\text{sign}(.6931(1) + 0.973(1) + 1.2827(-1)) = \text{positive}$

3. $\text{sign}(.6931(1) + 0.973(-1) + 1.2827(-1)) = \text{negative}$

4. $\text{sign}(.6931(-1) + 0.973(1) + 1.2827(1)) = \text{positive}$

5. $\text{sign}(.6931(-1) + 0.973(1) + 1.2827(-1)) = \text{negative}$

6. $\text{sign}(.6931(-1) + 0.973(-1) + 1.2827(-1)) = \text{negative}$

f) no misclassification error (error = 0) b/c Adaboost predicted them all correctly

		a	b	c	d	e	f	g	
2.	X	Y	Pred 1	e1	Pred 2	Prediction of y	e2	Pred 3	Final
	-1	1	$8/3$	$-5/3$	$-7/6$	$3/2$	-0.5	-0.5	1
	2	2	$8/3$	$-2/3$	$-7/6$	$3/2$	0.5	$1/4$	1.75
	4	5	$8/3$	$7/3$	$7/3$	5	0	$1/4$	5.25

a) first round Prediction = $\bar{y}$

$$\bar{y} = \frac{1+2+5}{3} = 8/3$$

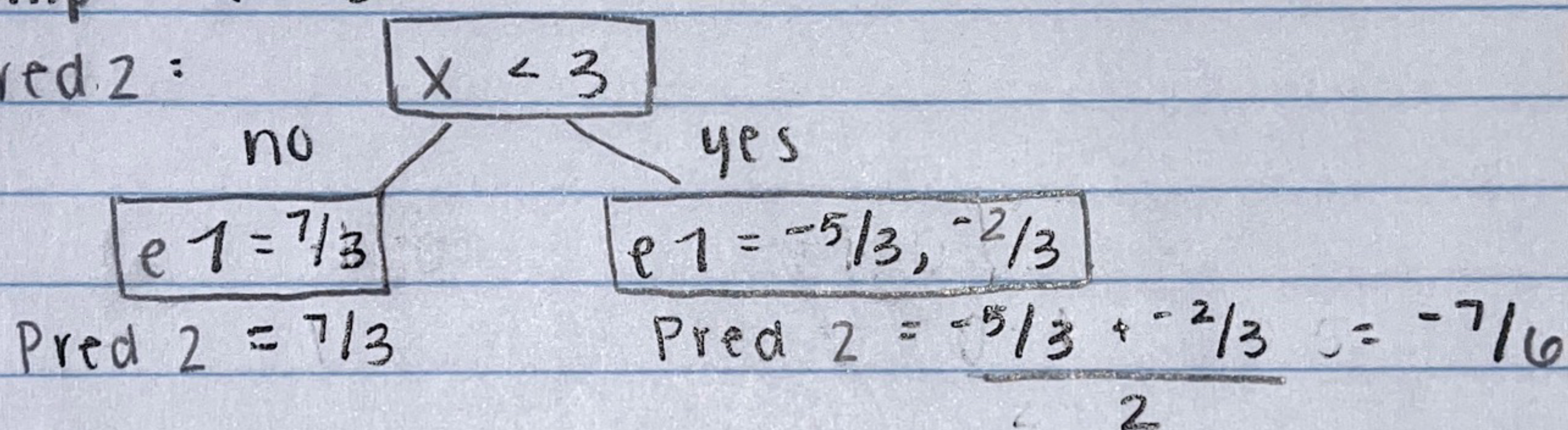
b) target of second round = errors of first round

$$e1 = Y - \text{Pred}_1$$

c) second round predicts errors of first round

stump: $X < 3$

Pred 2:



d) Prediction of y = (Pred 1 + Pred 2)

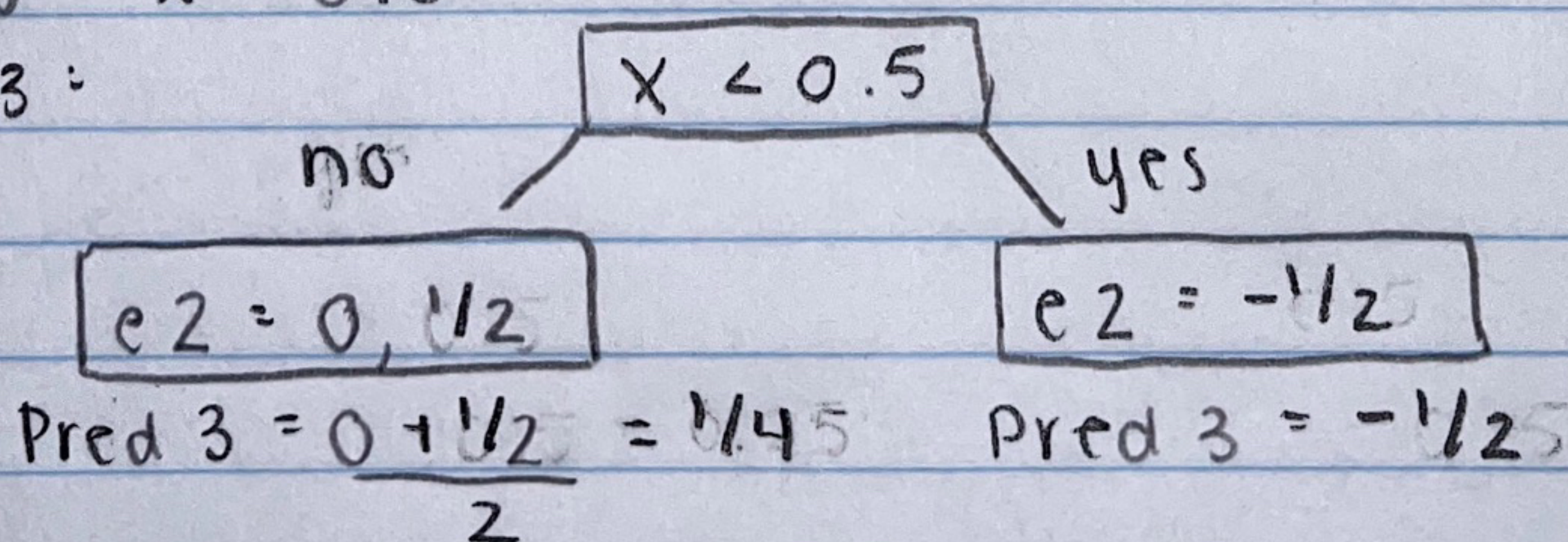
↑
final prediction

e) target of third round = $e2 = e1 - \text{Pred 2}$

f) third round predicts errors of second round

stump: $X < 0.5$

Pred 3:



g) final prediction for y after 3rd round:

$$\text{final_pred} = \text{Pred 1} + \text{Pred 2} + \text{Pred 3}$$

(see table above)